## 3.2 Groups

**Definition 3.2.1 (Group)**
*A **group** is a **monoid** $(G, \cdot)$ in which every element is invertible.*

*A group $G$ is said to be **abelian** if the binary operation is **commutative**. In this case we typically write the group operation additively*

$$g + h$$

**Definition 3.2.2 (Subgroup, Normal Subgroup)**
*A **subgroup** $H \leq G$ is a subset with the following properties*

- $e_G \in H$
- $x, y \in H \implies xy \in H$
- $x \in H \implies x^{-1} \in H$

*A subgroup $H$ is said to be **normal** in $G$ if in addition it satisfies*

$$gHg^{-1} \coloneqq \{ghg^{-1} \mid g \in G\} = H$$

*for all $g \in G$. NB it is easily verified that in an abelian group every subgroup is normal.*

**Proposition 3.2.3 (Subgroup is a group)**
*Let $H$ be a subgroup of $(G, \cdot)$ then $(H, \cdot|_{H \times H})$ is a group.*

**Example 3.2.4**
*$\mathbb{Z}$ is an abelian group under addition. The subgroups are of the form $n\mathbb{Z}$.*

**Definition 3.2.5**
*Let $(G, \cdot)$ and $(H, \cdot)$ be groups. A function $\varphi : G \to H$ is a **group homomorphism** if*

- $\varphi(e_G) = e_H$
- $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$

*Define the **image** of $\varphi$ to be*

$$\mathrm{Im}(\varphi) = \{\varphi(g) \mid g \in G\}$$

*and the **kernel** to be*

$$\ker(\varphi) \coloneqq \{g \in G \mid \varphi(g) = e_H\}$$

*It may be verified that $\mathrm{Im}(\varphi)$ is a subgroup of $H$ and $\ker(\varphi)$ is a normal subgroup of $G$.*

**Proposition 3.2.6 (Raise to the $n$-th power)**
*Let $g \in G$ be a group element. Then there exist a unique group homomorphism*

$$g^{(-)} : (\mathbb{Z}, +) \to (G, \cdot)$$

*satisfying*

$$g^1 = g$$

*In other words such that*

$$g^0 = e_G$$

$$g^{n+m} = g^n \cdot g^m \quad \forall n, m \in \mathbb{Z}$$

**Proposition 3.2.7**
*Let $g \in G$ be a group element. Then*

$$(g^n)^m = g^{nm}$$

*for all integers $n, m \in \mathbb{Z}$.*

**Definition 3.2.8 (Order of an element)**
*For $g \in G$ define the **order** of $g$ to be $o(g) \coloneqq \inf\{n \geq 0 \mid g^n = e\}$ where $\inf \emptyset = \infty$.*

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*We say $g$ has **finite order** if $o(g) \neq \infty$.*

**Definition 3.2.9 (Subgroup generated by an element)**
*The **subgroup generated** by an element $g \in G$ is defined to be $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$*

*It may be shown that when $g$ has finite order $n$ we have*

$$\langle g \rangle = \{e, g, ..., g^{n-1}\}$$

*and in particular $\#\langle g \rangle = o(g)$.*

**Proposition 3.2.10 (Cosets)**
*Let $H$ be a subgroup of $G$. The following is an equivalence relation on $G$*

$$g_1 \underset{H}{\sim} g_2 \iff g_1 g_2^{-1} \in H$$

*and the equivalence classes are precisely the sets of the form*

$$gH = \{gh \mid h \in H\} = [g]_{\underset{H}{\sim}}$$

*for some $g \in G$. Such an equivalence class is called a **right coset** and we denote the set of right cosets by*

$$G/H$$

*Define the **index** of $H$ in $G$ by $[G : H] := \#G/H$. When $H$ is finite each equivalence class has order $\#H$.*

*We say $\{g_i \in G\}_{i \in I}$ is a set of **coset representatives** for $H$ if the corresponding equivalence classes $\{[g_i]\}_{i \in I}$ are pairwise disjoint and cover $G$.*

*Proof.* It's trivial to show that $\underset{H}{\sim}$ is an equivalence relation (precisely because $H$ is a subgroup). Therefore by (2.1.6) the equivalence classes form a partition which we denote $G/H$.

We claim that $[g_1] = g_1 H$. Then $g_2 \in [g_1] \iff g_1 \underset{H}{\sim} g_2 \iff g_2 \underset{H}{\sim} g_1 \iff g_2 g_1^{-1} \in H \iff g_2 \in g_1 H$, which shows that the sets are equal.

The translation map $\psi_g : G \to G$ given by $g' \to gg'$ is bijective (for it has a two-sided inverse equal to $\psi_{g^{-1}}$). So in particular restricts to a bijective map $H \to gH$. This shows that all the cosets have the same order. ∎

**Example 3.2.11**
*$d\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ of index $d$. A set of coset representatives are $\{0, 1, ..., d-1\}$.*

**Corollary 3.2.12 (Lagrange's Theorem)**
*Let $H \leq G$ be a subgroup then*

$$\#G = [G : H] \times \#H$$

*More generally if $K \leq H$ then*

$$[G : K] = [G : H][H : K]$$

**Example 3.2.13**
*$d\mathbb{Z} \subseteq e\mathbb{Z} \iff d \mid e$ and $[e\mathbb{Z} : d\mathbb{Z}] = e/d$.*

**Proposition 3.2.14**
*Let $g \in G$ be an element of finite order. Then*

$$o(g) \mid \#G$$

*Furthermore*

$$g^n = e \iff o(g) \mid n$$

*Proof.* The first statement follows because the order $o(g)$ equals the order of the subgroup $\langle g \rangle$ generated by $g$.

Let $m = o(g)$ then by the division algorithm $n = qm + r$ for some $r < m$. Then $e = g^n = g^{qm} g^r = (g^m)^q g^r = e^q g^r = g^r$. By minimality we have $r = 0$ and $m \mid n$ as required. ∎

**Proposition 3.2.15 (Quotient Group)**
*Let $N$ be a normal subgroup $G$. Then the set of cosets*

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$$G/N$$

*forms a group under the binary operation*

$$g_1N \cdot g_2N \to (g_1g_2)N$$

*with identity $eN$.*

a) *There is a canonical surjective group homomorphism*

$$\pi : G \to G/N$$

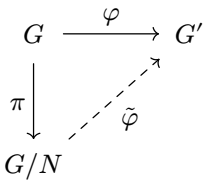
$$g \to gN$$

*with kernel $N$.*

b) *Let $N \subseteq H$ be a subgroup then define the correseponding subgroup of $G/N$*

$$H/N \coloneqq \pi(H) = \{hN \mid h \in H\}.$$

c) *Let $\varphi : G \to G'$ be a homomorphism with $N \subseteq \ker(\varphi)$, then there exists a unique homomorphism $\tilde{\varphi}$ making the diagram commute*




*such that*

a) $\mathrm{Im}(\varphi) = \mathrm{Im}(\tilde{\varphi})$
b) $\ker(\tilde{\varphi}) = \ker(\varphi)/N$

**Corollary 3.2.16 (Isomorphism Theorem)**
*Let $\varphi : G \to H$ be a group homomorphism, then there is a canonical isomorphism*

$$G/\ker(\varphi) \xrightarrow{\sim} \mathrm{Im}(\varphi)$$

**Proposition 3.2.17 (Correspondence Theorem)**
*Let $\pi : G \to G'$ be a surjective homomorphism with $\ker(\varphi) = N$ then there is a bijective correspondence of subgroups*

$$\begin{aligned} \{H \leq G \mid N \subseteq H\} &\longleftrightarrow \{H' \leq G'\} \\ H &\to \pi(H) \\ \pi^{-1}(H') &\leftarrow H' \end{aligned}$$

*which preserves index, that is*

$$[G' : H'] = [G : H]$$

*Furthermore $\#H' = [H : N]$.*

**Proposition 3.2.18**
*Suppose $N \subset G$ and $K \subset N$ are normal subgroups. Then $K \subset G$ is normal and there is a canonical isomorphism*

$$(G/N)/(K/N) \xrightarrow{\sim} G/K$$

*given by*

$$\overline{g+N} \to g + K$$

### 3.2.1 Cyclic Groups

**Definition 3.2.19**
*A group $G$ is said to be **cyclic** if there is a surjective group homomorphism*

$$(\mathbb{Z}, +) \longrightarrow (G, \cdot)$$

*equivalently if there is $g \in G$ such that $\langle g \rangle = G$. Such a $g$ is called a **generator** for $G$.*

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**Proposition 3.2.20**
*Consider the additive group $(\mathbb{Z}, +)$. Then*

a) *Every subgroup is of the form $d\mathbb{Z}$ for $d \geq 0$ and is itself cyclic*
b) *When $d > 0$, then $Z/d\mathbb{Z}$ has a complete set of coset representatives*

$$S := \{0, 1, ..., d-1\}$$

c) *In particular $[Z : d\mathbb{Z}] = d$ when $d > 0$*
d) *$d\mathbb{Z} \subseteq e\mathbb{Z} \iff e \mid d$ and in this case $[e\mathbb{Z} : d\mathbb{Z}] = \frac{d}{e}$*

*Proof.* We prove each in turn

a) By (2.2.6) every subgroup is of the form $d\mathbb{Z}$. Multiplication map $[d] : \mathbb{Z} \to d\mathbb{Z}$ shows it is itself cyclic.
b) By the division algorithm (2.2.5) $S$ is a complete set. Given $i, j \in S$ we note that $|i - j| < d$. And $i \underset{d}{\sim} j \implies d \mid |i - j| \implies |i - j| = 0 \implies i = j$. Therefore the set $S$ consists of distinct coset representatives.
c) This is clear from the previous step
d) The first equivalence is clear. By (3.2.12)

$$[\mathbb{Z} : d\mathbb{Z}] = [\mathbb{Z} : e\mathbb{Z}][e\mathbb{Z} : d\mathbb{Z}]$$

and the result follows.

■

**Proposition 3.2.21**
*Let $G$ be a cyclic group. Then*

- *If $G$ is infinite it is isomorphic to $\mathbb{Z}$*
- *If $G$ is finite it is isomorphic to $\mathbb{Z}/n\mathbb{Z}$ for some $n > 0$*

*Proof.* By the previous Proposition the kernel of the homomorphism $\mathbb{Z} \to G$ is of the form $n\mathbb{Z}$ for $n \geq 0$. By (...) $G$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. When $n = 0$ this is canonically isomorphic to $\mathbb{Z}$.

By the previous Proposition $\mathbb{Z}/n\mathbb{Z}$ is finite for $n > 0$ and therefore if $G$ is not finite we must have $n = 0$. Similarly If $G$ is finite then we must have $n > 0$. ■

We analyse the structure of finite cyclic groups in more detail. First recall the definition of **Euler's Totient Function**

**Definition 3.2.22 (Euler Totient Function)**
*Define the function*

$$\phi(n) = \#\{0 < d \leq n \mid (d, n) = 1\}$$

**Proposition 3.2.23 (Finite Cyclic Groups)**
*Consider a finite cyclic group $G$ of order $n$. Then*

a) *The order of $g^r$ is $\frac{n}{n,r}$ where $0 < r \leq n$.*
b) *There are $\phi(n)$ generators*
c) *For every $d \mid n$ there is a unique subgroup of order $\frac{n}{d}$ given by $\langle g^d \rangle$, which is cyclic.*
d) *For $d \mid n$ there are precisely $\phi(d)$ elements of order $d$*
e) *There are precisely $d$ elements of order dividing $d$*

*Proof.* We prove each in turn

a) $(g^r)^s = e_G \iff g^{rs} = e_G \overset{(3.2.14)}{\iff} n \mid rs \overset{(2.2.13)}{\iff} \frac{n}{n,r} \mid s$. Therefore $g^r$ has order $\frac{n}{n,r}$ as required.
b) Note $h$ is a generator iff $o(h) = n$. So $g^r$ is a generator iff $(n, r) = 1$ by the previous step. As $G = \{g, g^2, ..., g^n\}$ the result follows by definition of the totient function.
c) Recall there is a canonical surjective morphism $\pi : \mathbb{Z} \to G$ with kernel $n\mathbb{Z}$ and $\pi(1) = g$. By (3.2.17) the subgroups $H$ of $G$ correspond bijectively to subgroups $H'$ of $\mathbb{Z}$ containing $n\mathbb{Z}$, preserving the index. By (3.2.20) these are of the form $H' = d\mathbb{Z}$ for $d \mid n$, which correspond under $\pi$ to subgroups $H = \langle g^d \rangle$. Further $[G : \langle g^d \rangle] = [\mathbb{Z} : d\mathbb{Z}] = d$ whence $\#\langle g^d \rangle = \frac{n}{d}$. By definition $\langle g^d \rangle$ is cyclic.
d) Let $G[d]$ be the unique (cyclic) subgroup of order $d$. If $h$ has order $d$ then $\langle h \rangle$ has order $d$, and therefore by uniqueness is equal to $G[d]$. In particular $h \in G[d]$. Therefore by the previous part there are $\phi(d)$ elements of order $d$

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e) Suppose $h$ has order $e \mid d$. Both $G$ and $G[d]$ contain a unique subgroup of order $e$ and therefore by uniqueness this is simply $G[e] \subseteq G[d]$. Similarly by uniqueness $G[e] = \langle h \rangle$. Therefore $h \in G[d]$. Conversely suppose $h \in G[d]$ then $o(g) \mid d$ by (3.2.14). Therefore $G[d]$ consists of all the elements of order dividing $d$.

■

**Corollary 3.2.24**
*Let $n$ be a positive integer then*

$$n = \sum_{d|n} \phi(d)$$

*Proof.* Consider a cyclic group $G$ of order $n$. Every element has order dividing $n$ so the result follows from the previous Proposition by partitioning the group $G$ into subsets consisting of elements of equal order. ■

For an abelian group $G$ define the following subgroup

$$G[d] := \{g \in G \mid g^d = e\}.$$

We have shown for a cyclic group that $\#G[d] = d$ whenever $d \mid n$ and it is empty otherwise. We claim that this can be used to characterize cyclic groups. NB the following is adapted from this stackexchange answer

**Proposition 3.2.25 (Characterization of cyclic group)**
*Let $G$ be a finite abelian group such that $\#G[d] \leq d$ for all $d \mid n$. Then $G$ is cyclic.*

*Proof.* Let $n = \#G$ and $G_d$ be the subset of elements of order exactly $d$. Then we wish to show that $G_n$ is non-empty as any element of this set will be a generator. We actually show that $\#G_d = \phi(d) > 0$ whenever $d \mid n$.

Note that $G_d \subseteq G[d]$. If it's non-empty then for any $y \in G_d$, we have $\langle y \rangle$ is a subgroup of $G[d]$ of order $d$. As $\#G[d] \leq d$ we have $G[d] = \langle y \rangle$ is cyclic of order $d$. In other words $G_d$ is equal to the set of generators for $G[d]$. By the previous Proposition $G[d]$ has $\phi(d)$ generators. We conclude that for all $d \mid n$ we have $G_d$ is either empty or of order $\phi(d)$.

Therefore

$$n = \sum_{d|n} \#G_d \leq \sum_{d|n} \phi(d) = n$$

Therefore we must have equality everywhere and $\#G_d = \phi(d)$ as required. ■

**Example 3.2.26**
*Let $G = \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ of order $p^2$. Then $G[p] = G$ so $\#G[p] = p^2 > p$.*

## 3.2.2 Group Actions

**Definition 3.2.27 (Group Action)**
*Let $G$ be a group and $S$ a set. A group action of $G$ on $S$ is a map*

$$G \times S \longrightarrow S$$

$$(g, s) \longrightarrow g \cdot s$$

*such that*

- $es = s$
- $g(hs) = (gh)s$

**Definition 3.2.28 (Faithful group action)**
*A group action $G$ on $S$ is faithful if*

$$gs = s \quad \forall s \in S \Longrightarrow g = e$$

**Definition 3.2.29 (Free group action)**
*A group action $G$ on $S$ is free if*

$$g \neq e \Longrightarrow gs \neq s \quad \forall s$$

**Definition 3.2.30 (Orbit/Stabilizer)**
*Let $G$ be a group with an action on $S$ and $s \in S$. Define the stabilizer subgroup*

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$$G_s := \{g \in G \mid gs = s\}$$

*and the orbit*

$$Gs := \{gs \mid g \in G\}$$

**Proposition 3.2.31 ($S$ is disjoint union of orbits)**
*Let $G$ be a group with an action on $S$. Then the following is an equivalence relation*

$$s \sim t \iff gs = t \text{ some } g \in G$$

*and the equivalence classes are precisely the orbits of elements of $S$ under $G$. Further $S$ is the disjoint union of orbits.*

**Remark 3.2.32**
*An action is free if and only if $G_s = \{e\}$ for all $s \in S$.*

**Proposition 3.2.33 (Orbit-Stabilizer Theorem)**
*Let $G$ be a group with an action on $S$. Given an element $s \in S$ there is a natural bijection*

$$G/G_s \longrightarrow Gs$$

*between the cosets of $G_s$ and the orbit $Gs$. In particular when $G$ is finite*

$$\#G = \#(Gs) \times \#G_s$$

*and when the action is free*

$$\#G = \#(Gs)$$

### 3.2.3 Symmetric Group

**Definition 3.2.34 (Symmetric Group)**
*Let $S_n$ denote the set of permutations (bijections) of $\{1, ..., n\}$.*

*Permutations $\sigma, \tau \in S_n$ are called **disjoint** if the supports are disjoint. Note disjoint permutations commute.*

**Definition 3.2.35 (Cycle)**
*Let $i_1, ..., i_r \in J_n$ be an ordered $r$-tuple, the permutation which maps*

$$i_k \to \begin{cases} i_{k+1} & k < r \\ i_1 & k = r \end{cases}$$

*is denoted by $(i_1\, i_2 \ldots i_r)$ and called a **cycle**.*

*A cycle with two elements $(i\, j)$ is called a **transposition**. Finally an **adjacent transposition** is one of the form $(i\, i+1)$.*

**Proposition 3.2.36**
*Let $\sigma \in S_n$. Then $\sigma$ may be represented by*

*a) a product of disjoint cycles, which is unique up to permutation of cycles.*
*b) a product of transpositions, the number of which is unique modulo 2*
*c) There is a well-defined group homomorphism*

$$\varepsilon : S_n \to \{-1, 1\}$$

*such that*

$$\varepsilon((i\, j)) = -1$$

*and $\varepsilon(1) = 1$.*
*d) A cycle $\sigma$ of length $r$ satisfies $\varepsilon(\sigma) = (-1)^{r+1}$*

**Proposition 3.2.37 (Adjacent Transpositions Generate the Symmetric Group)**
*Every $\sigma \in S_n$ is a product of adjacent transpositions $(i\, i+1)$, $1 \le i < n$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Every permutation is a product of transpositions, so it suffices to write each transposition $(i\, j)$ with $i < j$ as a product of adjacent ones, which we do by induction on $j - i$. If $j - i = 1$ there is nothing to show. If $j - i > 1$ then

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$$(i\,j) = (j-1\,j)(i\,j-1)(j-1\,j)$$

since conjugating $(i\,j-1)$ by $(j-1\,j)$ replaces $j-1$ with $j$, and $(i\,j-1)$ is a product of adjacent transpositions by induction. ∎

### 3.2.4 Shuffle Permutations

**Definition 3.2.38**
*Let $p+q=n$ be positive integers. We define the subgroup of* ***shuffle permutations***

$$\mathrm{Sh}(p,q) \coloneqq \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q)\}$$

*Similarly if $p+q+r=n$ are positive integers define the subgroup*

$$\mathrm{Sh}(p,q,r) \coloneqq \{\sigma \in S_n \mid \sigma(1) < ... < \sigma(p) \text{ and } \sigma(p+1) < ... < \sigma(p+q) \text{ and } \sigma(p+q+1) < ... < \sigma(p+q+r)\}$$

*Let $u : [1,p] \to [1,n]$ and $v : [1,q] \to [1,n]$ be injective maps such that $u([1,p]) \cap v([1,q]) = \emptyset$. Define $u \star v \in S_n$ by*

$$(u \star v)(i) \coloneqq \begin{cases} u(i) & 1 \le i \le p \\ v(i-p) & p < i \le p+q \end{cases}$$

*If $u, v$ are order preserving then $u \star v \in \mathrm{Sh}(p,q)$. Every shuffle permutation is of this form.*

**Lemma 3.2.39**
*Let $X, Y \subset \mathbb{N}$ be a finite sets of order $n$. There exists a unique order isomorphism*

$$u : X \to Y$$

**Proposition 3.2.40**
*Let $p+q=n$ be positive integers. There is a bijection*

$$\mathrm{Sh}(p,q) \to \{u : [1,p] \to [1,n] \text{ order preserinvg}\}$$

$$\sigma \to \sigma|_{[1,p]}$$

*Proof.* We provide an explicit inverse. For given $u$ the set $Y \coloneqq [1,n] \setminus u([1,p])$. As $u$ is order preserving it is injective, whence $\#Y = n - \#u([1,p]) = n-p = q$. Therefore there is an order isomorphism $v : [1,q] \to Y$ and we may define the shuffle permutation $\sigma \coloneqq u \star v$. ∎

**Proposition 3.2.41**
*Let $p+q=n$ be positive integers. There is an injective group homomorphism*

$$\begin{aligned} S_p \times S_q &\to S_n \\ (\sigma_1, \sigma_2) &\to \sigma_1 \star (\sigma_2 + p) \end{aligned}$$

*The image is the set $\sigma \in S_n$ for which $\sigma([1,p]) = [1,p]$ (equivalently $\sigma([p+1,q]) = \sigma([p+1,q])$.*

**Proposition 3.2.42**
*Let $p+q=n$ be positive integers. There is a bijection*

$$\mathrm{Sh}(p,q) \to S_n / (S_p \times S_q)$$

*Furthermore $\#\,\mathrm{Sh}(p,q) = \frac{(p+q)!}{p!q!}$. By a similar argument $\#\,\mathrm{Sh}(p,q,r) = \frac{(p+q+r)!}{p!q!r!}$.*

*Proof.* Let $\sigma \in S_n$. Then there exists an order isomorphism $u : [1,p] \to \sigma([1,p])$ and $v : [1,q] \to \sigma([p+1,p+q])$. Define $\sigma_1(i) \coloneqq u^{-1}(\sigma(i))$ and $\sigma_2(j) \coloneqq v^{-1}(\sigma(j+p))$. Then we may verify that $\sigma_1 \in S_p$, $\sigma_2 \in S_q$ and

$$(u \star v) \circ (\sigma_1 \star (\sigma_2 + p)) = \sigma$$

so the map is surjective.

Suppose $\sigma, \sigma' \in \mathrm{Sh}(p,q)$ are such that

$$\sigma = \sigma' \circ (\sigma_1 \star (\sigma_2 + p))$$

for $\sigma_1 \in S_p$ and $\sigma_2 \in S_q$. If $\sigma_1 \neq e$ then there exists $1 \le i < j \le p$ such that $\sigma_1(j) < \sigma_1(i)$. Then by monotonicity of $\sigma'$ we have $\sigma(j) = \sigma'(\sigma_1(j)) < \sigma'(\sigma_1(i)) = \sigma(i)$, which contradicts monotonicity of $\sigma$. So we conclude $\sigma_1 = e$ and similarly $\sigma_2 = e$. Therefore $\sigma = \sigma'$, and the map is injective. By the Orbit-Stabilizer theorem we have

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$$\#\operatorname{Sh}(p,q) = \#S_n / \#\big(S_p \times S_q\big)$$

from which the result follows. ∎

**Definition 3.2.43 (Circulant Permutation)**
*Let $n$ be an integer, and write $\rho_n \in S_n$ for the permutation*

$$\{1,...,n\} \to \{2,3,...,n,n+1\}$$

*Note that $\varepsilon(\rho_n) = (-1)^{n+1}$.*

**Proposition 3.2.44 (Symmetry of Shuffling)**
*Let $p+q=n$ be positive integers. Then there is a bijection*

$$\begin{aligned} \operatorname{Sh}(p,q) &\to \operatorname{Sh}(q,p) \\ \sigma &\to \sigma \circ \rho^p_{p+q} \end{aligned}$$

*Note that $\varepsilon\big(\rho^p_{p+q}\big) = (-1)^{pq}$ by (3.2.43).*

**Proposition 3.2.45 (Associativity of Shuffling)**
*The following map is a bijection*

$$\begin{aligned} \pi : \operatorname{Sh}(p+q,r) \times \operatorname{Sh}(p,q) &\to \operatorname{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ (\sigma_2 \star (1_r + p + q)) \end{aligned}$$

*Similarly so is*

$$\begin{aligned} \pi' : \operatorname{Sh}(p,q+r) \times \operatorname{Sh}(q,r) &\to \operatorname{Sh}(p,q,r) \\ (\sigma_1,\sigma_2) &\to \sigma_1 \circ \big(1_p \star (\sigma_2 + p)\big) \end{aligned}$$

*Proof.* It is straightforward to verify that $\pi(\sigma_1,\sigma_2)$ is injective and satisfies the ordering properties. Therefore $\pi$ is well-defined and we claim it is surjective.

Given $\sigma \in \operatorname{Sh}(p,q,r)$ let $u : [1,p+q] \to \sigma([1,p+q])$, $v : [1,r] \to \sigma([p+q+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2 := u^{-1} \circ \sigma|_{[1,p+q]}$. We may verify that $\sigma_1 \in \operatorname{Sh}(p+q,r)$ and $\sigma_2 \in \operatorname{Sh}(p,q)$, and furthermore that $\pi(\sigma_1,\sigma_2) = \sigma$. By counting $\pi$ is injective.

For the second case let $u : [1,p] \to \sigma([1,p])$, $v : [1,q+r] \to \sigma([p+1,p+q+r])$ be the unique order isomorphisms. Define $\sigma_1 = u \star v$ and $\sigma_2(j) := v^{-1}(\sigma(j+p))$. The verification follows as before. ∎

## 3.2.5 Totally Ordered Abelian Group

**Definition 3.2.46 (Ordered Abelian Group)**
*An abelian group $(G,+)$ together with a total order $\leq$ is an **ordered abelian group** if it satisfies*

$$x \leq y \implies x + z \leq y + z \quad \forall x,y,z \in G$$

*Define $G^+ := \{g \in G \mid (0 \leq g) \wedge (g \neq 0)\}$ and $G^- := \{g \in G \mid (g \leq 0) \wedge (g \neq 0)\}$.*

## 3.2.6 Sums in an Abelian Group

**Proposition 3.2.47 (Finite Sums in an Abelian Group)**
*Let $G$ be an abelian group. Define $\psi_0 := 0 \in G$ (the value on the one-point set $G^0$) and, for $n \geq 1$, maps $\psi_n : G^n \to G$ recursively by*

$$\psi_n(x_1,...,x_n) = x_1 + \psi_{n-1}(x_2,...,x_n)$$

*so that in particular $\psi_1 = 1_G$. Then each $\psi_n$ is invariant under permutations, that is $\psi_n\big(x_{\sigma(1)},...,x_{\sigma(n)}\big) = \psi_n(x_1,...,x_n)$ for every $\sigma \in S_n$. Moreover:*

*a) $\psi_{m+r}(x_1,...,x_m,y_1,...,y_r) = \psi_m(x_1,...,x_m) + \psi_r(y_1,...,y_r)$ for $m,r \geq 0$*
*b) $\psi_n(0,...,0) = 0$ and $\psi_{n+1}(x_1,...,x_n,0) = \psi_n(x_1,...,x_n)$ for $n \geq 0$*
*c) $\psi_n(x_1+y_1,...,x_n+y_n) = \psi_n(x_1,...,x_n) + \psi_n(y_1,...,y_n)$ for $n \geq 0$*
*d) if $\varphi : G \to H$ is a group homomorphism to an abelian group $H$ then $\psi_n(\varphi(x_1),...,\varphi(x_n)) = \varphi(\psi_n(x_1,...,x_n))$*
*e) $\psi_n(-x_1,...,-x_n) = -\psi_n(x_1,...,x_n)$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

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We show that each $\psi_n$ is invariant under permutations by induction on $n$; for $n \le 1$ there is nothing to show. Since $S_n$ is generated by the adjacent transpositions ((3.2.37)) it suffices to check invariance under $(i\,i+1)$ for $1 \le i < n$, and let $n \ge 2$ with $\psi_{n-1}$ invariant.

If $i \ge 2$ the transposition only exchanges two entries of $(x_2, ..., x_n)$, so $\psi_n$ is unchanged since $\psi_{n-1}$ is invariant. If $i = 1$ then

$$\begin{aligned}\psi_n(x_2, x_1, x_3, ..., x_n) &= x_2 + (x_1 + \psi_{n-2}(x_3, ..., x_n)) && \text{by the recursion}\\ &= x_1 + (x_2 + \psi_{n-2}(x_3, ..., x_n)) && \text{by associativity and commutativity in } G\\ &= \psi_n(x_1, x_2, x_3, ..., x_n) && \text{by the recursion}\end{aligned}$$

For a) we use induction on $m$. For $m = 0$ both sides are $\psi_r(y_1, ..., y_r)$, as $\psi_0 = 0$. For $m \ge 1$, by the recursion, the induction hypothesis and associativity,

$$\psi_{m+r}(x_1, ..., x_m, y_1, ..., y_r) = x_1 + (\psi_{m-1}(x_2, ..., x_m) + \psi_r(y_1, ..., y_r)) = \psi_m(x_1, ..., x_m) + \psi_r(y_1, ..., y_r)$$

Next, $\psi_n(0, ..., 0) = 0 + \psi_{n-1}(0, ..., 0) = 0$ by induction on $n$, and by a) with $r = 1$ (so $\psi_1(0) = 0$) $\psi_{n+1}(x_1, ..., x_n, 0) = \psi_n(x_1, ..., x_n) + 0$, which proves b). The remaining properties are proved by induction on $n$, the case $n = 0$ being clear ($\psi_0 = 0$, and $\varphi(0) = 0$ for d)). For c), writing $x' = (x_2, ..., x_n)$ and $y' = (y_2, ..., y_n)$,

$$\psi_n(x + y) = (x_1 + y_1) + \psi_{n-1}(x' + y') = (x_1 + \psi_{n-1}(x')) + (y_1 + \psi_{n-1}(y')) = \psi_n(x) + \psi_n(y)$$

using the induction hypothesis and commutativity. For d), $\psi_n(\varphi(x)) = \varphi(x_1) + \psi_{n-1}(\varphi(x')) = \varphi(x_1) + \varphi(\psi_{n-1}(x')) = \varphi(\psi_n(x))$ since $\varphi$ is additive. Finally e) is d) for the homomorphism $x \mapsto -x$ of $G$. ∎

**Definition 3.2.48 (Sum of a Finite Family)**
*Let $G$ be an abelian group, written additively. For a family $(g_i)_{i \in I}$ in $G$ indexed by a finite set $I$ with $n \ge 0$ elements, define*

$$\sum_{i \in I} g_i := \psi_n\left(g_{e(1)}, ..., g_{e(n)}\right)$$

*for any enumeration of $I$, that is a bijection $e : \{1, ..., n\} \to I$, where $\psi_n$ is as in (3.2.47). In particular the sum over the empty set is $\psi_0 = 0$. If $g_i = 0$ for all $i \in I$ then $\sum_{i \in I} g_i = 0$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

If $e, e' : \{1, ..., n\} \to I$ are two enumerations of $I$ then $\sigma := e^{-1} \circ e' \in S_n$ and $g_{e'(k)} = g_{e(\sigma(k))}$, so the sum is independent of the enumeration since $\psi_n$ is invariant under permutations by (3.2.47). If $g_i = 0$ for all $i \in I$ then every entry of $\psi_n\left(g_{e(1)}, ..., g_{e(n)}\right)$ is 0, so the sum is 0 since $\psi_n(0, ..., 0) = 0$ by the same proposition. ∎

**Proposition 3.2.49 (Sum over a Disjoint Union)**
*Let $G$ be an abelian group and $(g_i)_{i \in I \cup J}$ a family in $G$ indexed by the union of two disjoint finite sets $I$ and $J$. Then*

$$\sum_{i \in I \cup J} g_i = \sum_{i \in I} g_i + \sum_{i \in J} g_i$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $|I| = m \ge 0$ and $|J| = r \ge 0$, and let $e_I : \{1, ..., m\} \to I$ and $e_J : \{1, ..., r\} \to J$ be enumerations. Since $I$ and $J$ are disjoint, $e : \{1, ..., m + r\} \to I \cup J$ with $e(k) = e_I(k)$ for $k \le m$ and $e(k) = e_J(k - m)$ for $k > m$ is an enumeration of $I \cup J$, and by (3.2.47) (the identity for $\psi_{m+r}$) and independence of the enumeration ((3.2.48)) $\sum_{i \in I \cup J} g_i = \psi_{m+r}\left(g_{e(1)}, ..., g_{e(m+r)}\right) = \psi_m\left(g_{e_I(1)}, ..., g_{e_I(m)}\right) + \psi_r\left(g_{e_J(1)}, ..., g_{e_J(r)}\right) = \sum_{i \in I} g_i + \sum_{i \in J} g_i$. ∎

**Definition 3.2.50 (Sum of an Almost Zero Family)**
*Let $G$ be an abelian group, written additively. Let $(g_i)_{i \in I}$ be a family in $G$ indexed by an arbitrary set $I$ for which all but finitely many $g_i$ are zero, that is whose* ***support*** $\operatorname{supp}(g) := \{i \in I \mid g_i \neq 0\}$ *is finite. Define*

$$\sum_{i \in I} g_i := \sum_{i \in \operatorname{supp}(g)} g_i$$

*using the sum over the finite set $\operatorname{supp}(g)$ from (3.2.48). Then $\sum_{i \in I} g_i = \sum_{i \in J} g_i$ for every finite subset $J \subseteq I$ containing $\operatorname{supp}(g)$. In particular, if $I$ is finite this is the sum of the finite family $(g_i)_{i \in I}$ from (3.2.48) (take $J = I$).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

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Let $J \subseteq I$ be finite with $\operatorname{supp}(g) \subseteq J$. Then $J$ is the disjoint union of $\operatorname{supp}(g)$ and $J \setminus \operatorname{supp}(g)$, and $g_i = 0$ for every $i \in J \setminus \operatorname{supp}(g)$, so by (3.2.49) and (3.2.48) (a sum of zero terms is 0)

$$\sum_{i \in J} g_i = \sum_{i \in \operatorname{supp}(g)} g_i + \sum_{i \in J \setminus \operatorname{supp}(g)} g_i = \sum_{i \in \operatorname{supp}(g)} g_i$$

which is the defining value. If $I$ is finite we may take $J = I$. ∎

**Proposition 3.2.51 (Sums of Almost Zero Families)**
*Let $G$ be an abelian group and $(g_i)_{i \in I}$, $(h_i)_{i \in I}$ families in $G$ indexed by the same set $I$, with all but finitely many terms zero. Then $(g_i + h_i)_{i \in I}$ and $(-g_i)_{i \in I}$ are again families with all but finitely many terms zero, and*

*a) $\sum_{i \in I}(g_i + h_i) = \sum_{i \in I} g_i + \sum_{i \in I} h_i$*
*b) $\sum_{i \in I}(-g_i) = -\sum_{i \in I} g_i$, and consequently $\sum_{i \in I}(g_i - h_i) = \sum_{i \in I} g_i - \sum_{i \in I} h_i$*
*c) if $\varphi : G \to H$ is a group homomorphism to an abelian group $H$ (for instance multiplication by an element $r$ of a ring $R$ on the additive group $G = R$) then $(\varphi(g_i))_{i \in I}$ again has all but finitely many terms zero, and $\sum_{i \in I} \varphi(g_i) = \varphi\left(\sum_{i \in I} g_i\right)$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $J \subseteq I$ be a finite subset containing every $i$ with $g_i \neq 0$ or $h_i \neq 0$. For $i \notin J$ we have $g_i + h_i = 0$, $-g_i = 0$ and $\varphi(g_i) = \varphi(0) = 0$, so all three families have all but finitely many terms zero, and $J$ is a valid choice of finite subset in the definition of all of the sums involved. Therefore

$$\sum_{i \in I}(g_i + h_i) = \sum_{i \in J}(g_i + h_i) = \sum_{i \in J} g_i + \sum_{i \in J} h_i = \sum_{i \in I} g_i + \sum_{i \in I} h_i$$

$$\sum_{i \in I}(-g_i) = \sum_{i \in J}(-g_i) = -\sum_{i \in J} g_i = -\sum_{i \in I} g_i$$

$$\sum_{i \in I} \varphi(g_i) = \sum_{i \in J} \varphi(g_i) = \varphi\left(\sum_{i \in J} g_i\right) = \varphi\left(\sum_{i \in I} g_i\right)$$

using that finite sums are additive and commute with negation and with group homomorphisms ((3.2.47)). Finally $g_i - h_i = g_i + (-h_i)$, so the last claim of b) follows from a) and the first part of b). ∎

**Proposition 3.2.52 (Sums over a Partition)**
*Let $G$ be an abelian group and $(g_i)_{i \in I}$ a family in $G$ with all but finitely many terms zero. If $I = \bigsqcup_{\lambda \in \Lambda} I_\lambda$ is a partition of $I$ into disjoint subsets then each $(g_i)_{i \in I_\lambda}$ and the family $\left(\sum_{i \in I_\lambda} g_i\right)_{\lambda \in \Lambda}$ again have all but finitely many terms zero, and*

$$\sum_{i \in I} g_i = \sum_{\lambda \in \Lambda} \sum_{i \in I_\lambda} g_i$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $J \subseteq I$ be finite containing every $i$ with $g_i \neq 0$, and let $\Lambda_J := \{\lambda \in \Lambda \mid I_\lambda \cap J \neq \emptyset\}$, which is finite. For $\lambda \in \Lambda_J$ the set $J \cap I_\lambda$ is a valid choice for the subfamily, so $\sum_{i \in I_\lambda} g_i = \sum_{i \in J \cap I_\lambda} g_i$, while for $\lambda \notin \Lambda_J$ all terms of the subfamily are zero and $\sum_{i \in I_\lambda} g_i = 0$. So $\left(\sum_{i \in I_\lambda} g_i\right)_{\lambda \in \Lambda}$ is almost zero with $\Lambda_J$ a valid choice, and since $J$ is the disjoint union of the finitely many $J \cap I_\lambda$, $\lambda \in \Lambda_J$, induction on $|\Lambda_J|$ using (3.2.49) gives

$$\sum_{i \in I} g_i = \sum_{i \in J} g_i = \sum_{\lambda \in \Lambda_J} \sum_{i \in J \cap I_\lambda} g_i = \sum_{\lambda \in \Lambda} \sum_{i \in I_\lambda} g_i$$

∎

**Lemma 3.2.53 (Reindexing and Iterated Sums)**
*Let $G$ be an abelian group.*

*a) If $(g_i)_{i \in I}$ has all but finitely many terms zero and $\sigma : I' \to I$ is a bijection, then $\left(g_{\sigma(i')}\right)_{i' \in I'}$ has all but finitely many terms zero and $\sum_{i' \in I'} g_{\sigma(i')} = \sum_{i \in I} g_i$.*
*b) If $\left(g_{(i,k)}\right)_{(i,k) \in I \times K}$ has all but finitely many terms zero then so do each row $\left(g_{(i,k)}\right)_{i \in I}$ and $\left(\sum_{i \in I} g_{(i,k)}\right)_{k \in K}$, and*

$$\sum_{(i,k) \in I \times K} g_{(i,k)} = \sum_{k \in K} \sum_{i \in I} g_{(i,k)}$$

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*and likewise with the roles of $i$ and $k$ exchanged.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

a) Let $J \subseteq I$ be finite containing every $i$ with $g_i \neq 0$. Then $\sigma^{-1}(J)$ is finite and contains every $i'$ with $g_{\sigma(i')} \neq 0$, so the reindexed family has all but finitely many terms zero, and

$$\sum_{i' \in I'} g_{\sigma(i')} = \sum_{i' \in \sigma^{-1}(J)} g_{\sigma(i')} = \sum_{i \in J} g_i = \sum_{i \in I} g_i$$

since $\sigma$ restricts to a bijection $\sigma^{-1}(J) \to J$ and finite sums do not depend on the ordering of the terms.

b) Apply (3.2.52) to the partition $I \times K = \bigsqcup_{k \in K} I \times \{k\}$. The block $\left(g_{(i,k')}\right)_{(i,k') \in I \times \{k\}}$ is the reindexing of the row $\left(g_{(i,k)}\right)_{i \in I}$ along the bijection $i \mapsto (i,k)$, so by a) the rows have all but finitely many terms zero, with $\sum_{(i,k') \in I \times \{k\}} g_{(i,k')} = \sum_{i \in I} g_{(i,k)}$. Hence the partition formula reads $\sum_{(i,k) \in I \times K} g_{(i,k)} = \sum_{k \in K} \sum_{i \in I} g_{(i,k)}$, and $\left(\sum_{i \in I} g_{(i,k)}\right)_{k \in K}$ has all but finitely many terms zero. The other order is the same argument applied to the partition $I \times K = \bigsqcup_{i \in I} \{i\} \times K$. ∎

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